



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade Awarded Category - I by UGC

Seat No.							
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Institute: (0203)SYMBIOSIS CENTRE FOR MANAGEMENT AND HUMAN RESOURCE DEVELOPMENT, PUNE

Programme: (020341) MASTER OF BUSINESS ADMINISTRATION

Batch: 2024-26

Semester: I

Course: Business Statistics

Course Code: 0203410102

Date: 10/03/2025

Maximum Marks: 40

Day: Monday

Time: 09:30 - 11:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instruction:

- 1) All questions are compulsory.

Q1. A company claims that the weight of their bags follows a normal distribution with a mean of 50 kg and a standard deviation of 4 kg.

- a. What is the probability that a randomly selected bag weighs more than 54 kg?
- b. What is the probability that a randomly selected bag weighs less than 46 kg?
- c. What is the probability that a randomly selected bag weighs between 48 kg and 52 kg?

Also write a short note on Normal Distribution

(CO-1) (10 Marks)

Q2 A retail chain claims that the average customer satisfaction score is 75 out of 100. To test this claim, a sample of 30 customers is surveyed, and their average satisfaction score is found to be 78 with a standard deviation of 10.

At a 1% significance level, can you conclude that the average satisfaction score is different from 75? Also write a short note on t test

(CO-2) (10 Marks)

Q3. A sales manager wants to determine if sales performance differs among four regions: North, South, East, and West. The manager collects sales performance scores (in thousands) from 20 employees, with 5 employees from each region. The scores are as follows:

- **North:** 120, 125, 130, 115, 140
- **South:** 110, 105, 120, 125, 130
- **East:** 140, 150, 145, 155, 160
- **West:** 100, 95, 105, 110, 120

At a 5% significance level, can you conclude that sales performance differs among the four regions?

create null and alternate hypotheses, and suggest strategies for improving sales performance.
Explain one way ANOVA

(CO-3) (10 Marks)

Q4. A company wants to determine if there is an association between education level and job satisfaction. The following data is collected from 200 employees:

Education Level	Satisfied	Not Satisfied	Total
High School	50	30	80
Bachelor's	60	40	100
Master's	25	25	50
Total	135	95	200

At a 5% significance level, can you conclude that job satisfaction is associated with education level?

(CO-4) (10 Marks)
